KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering
Summer-2023 Mid-term Examination
Program: B.Sc. in CSE (Batch – 16^{th}) 1^{st} Year 1^{st} Semester
Course Title: Physics - I
Course Code: CSE 0533 1101

PART-A: MCQ

Time: 10 Minutes **Total Marks: 10x1=10**

Student's ID: | | | | | | | | | | | | | | | |

Date: | | | | | | | | **Invigilator's Signature:**

(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

Cognitive Knowledge

1. What is the value of the minimum amount of charge in universe?
A) $1.6 \times 10^{-16} C$
B) $1.6 \times 10^{-19} C$
C) $9.1 \times 10^{-31} C$
D) $1.9 \times 10^{-16} C$
E) $3.1 \times 10^{-34} C$

Cognitive Knowledge

2. The tangent of any point in the electric field lines indicates the …………….
A) current.
B) voltage.
C) emf.
D) resistance.
E) electric field intensity.

Cognitive Understanding

3. What is the value of the proportionality constant (in TmA^{-1}) of Biot-Savart Law?
A) 10^{-7}
B) $2\pi \times 10^{-7}$
C) $4\pi \times 10^{-7}$
D) 2π
E) 4π

Cognitive Knowledge

4. Which is the unit of electric potential energy?
A) Am^{-1}
B) $Csec^{-1}$
C) Fm^{-1}
D) $Joule$
E) VA^{-1}

Cognitive Understanding

5. What will be the magnitude of the magnetic field inside the wire that is produced by itself?
A) 0
B) 1
C) 2π
D) 4π
E) Infinite

Cognitive Application

6. What is the permittivity of the medium of dielectric constant 1.5?
   A) $1.5 \times 10^{-12} Fm^{-1}$
   B) $8.85 \times 10^{-12} Fm^{-1}$
   C) $13.3 \times 10^{-12} Fm^{-1}$
   D) $15.5 \times 10^{-12} Fm^{-1}$
   E) $20.1 \times 10^{-12} Fm^{-1}$

Cognitive Understanding

7. To get the maximum flux, what will be the angle between $\bar{E}$ and $\overline{da}$?
   A) -30^0
   B) 0^0
   C) 45^0
   D) 90^0.
   E) 120^0.

Cognitive Application

8. What is the linear charge density of the wire of length 200cm having 1000C charge?
   A) 500C/m
   B) 50C/m
   C) 5C/m
   D) 2C/m
   E) 0.5C/m

Cognitive Understanding

9. If we double the length of a conducting wire, then what will be the specific resistance of that wire?
   A) Triple
   B) Double
   C) Half
   D) Quarter
   E) Same

Cognitive Knowledge

10. Drift velocity is …….. the current density.
    A) proportional to
    B) inversely proportional to
    C) greater than
    D) less than
    E) equal to

Signature of Examiner

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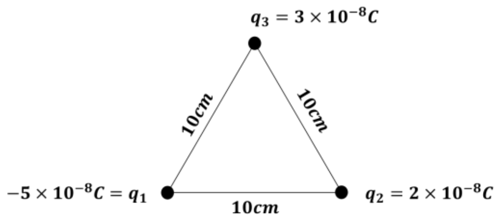
PART-B: SAQ

Time: 1 Hour and 20 Minutes **Total Marks: 4x5=20**

INSTRUCTIONS: Answer question no. 01 (one) and any 03 (three) from the rest. Marks are shown by side.

Question No. Marks

1. See the stem carefully and answer the question that follows:

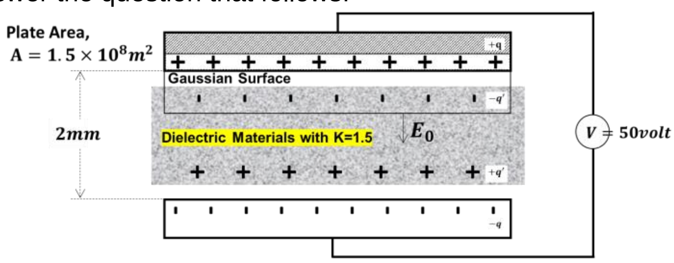


a) Define 1 volt. [1]
b) Find the electric potential energy for the above system. [2]
c) Illustrate if there is no q_1 charge in the above system then what will be the structure of electric lines of force? [2]

2. a) Define charge density. [1]
b) Relate drift velocity with charge density. [2]
c) A silver wire 1 mm in diameter carries a charge of 90 C in 1 hr. Silver contains 5.8×10^{28} free electrons/m^3. Calculate – [2]
(i) the current density of the wire.
(ii) the drift velocity of electrons in the wire.

3. a) Write down the properties of electric lines of forces. [1]
b) Mention the precautionary steps to stay safe from lightning. [2]
c) A negative charge of 10^{-6} is situated in air at the origin of a rectangular co-ordinate system. A second negative point charge of 10^{-4} is situated on the positive X-axis at a distance of 50 cm from the origin. What is the force on the second charge? [2]

4. See the stem and answer the question that follows:



In Physics class, a student 'Asif' said that "Free Charge, q < Induced Charge, q'"

a) Specify K. [1]
b) Calculate the capacitance of the mentioned capacitor. [2]
c) Is it true Asif's statement? If not, give your mathematical logic [2]

5. a) Define 1 Tesla. [1]
b) Conclude Oersted experiment. [1]
c) An electron is moving with a speed of $5 \times 10^7 \mathrm{m/s}$ at right angle to a magnetic field of $5000\mathrm{G}$. (i) What is the magnetic force on the electron? (ii) What is the radius of circle in which the electron moves? [3]